

Frequency Modulation and Demodulation

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Abstract—This project presents a compact and cost-effective approach to frequency modulation (FM) and demodulation using the CD4046 phase-locked loop (PLL) integrated circuit. The system employs the IC's voltage-controlled oscillator (VCO) to achieve FM by varying the carrier frequency in response to the amplitude of the input signal. Demodulation is performed using the phase comparator mode of the same IC, which recovers the original modulating signal by detecting phase differences between the received signal and a reference. Experimental results validate the design's effectiveness in signal transmission, demonstrating its suitability for practical communication applications.

I. INTRODUCTION

Frequency modulation is a process of encoding information on a signal (analogue or digital) by varying the carrier wave frequency in accordance with the amplitude of the modulating signal. The FM signal can be expressed by the following equation:

$$x_{FM}(t) = A_c \cos(\theta(t))$$

$$= A_c \cos\left(2\pi f_c t + 2\pi k_f \int_0^t x(\lambda) d\lambda\right) \quad (1)$$

If

$$x(\lambda) = A_m \cos(2\pi f_m \lambda), \quad (2)$$

then:

$$x_{FM}(t) = A_c \cos(2\pi f_c t + \beta \sin(2\pi f_m t)) \quad (3)$$

Where

$\theta(t)$ = instantaneous modulated frequency

f_c = carrier frequency

k_f = frequency sensitivity

f_m = modulating frequency

$$\beta = \text{modulation index} = \frac{k_f A_m}{f_m} = \frac{\Delta f}{f_m}$$

The frequency of FM signal may be expressed as:

$$f = \frac{1}{2\pi} \frac{d}{dt} (\theta(t)) = f_c + f_m \beta \cos(2\pi f_m t) \quad (4)$$

Equation (4) shows that the instantaneous frequency of the FM signal increases or decreases relative to the carrier's center frequency depending on the amplitude of the modulating signal. In this project, both FM modulation and demodulation were done using the IC CD4046. The CD4046 is a versatile integrated circuit that contains both a Phase-Locked Loop (PLL) and a Voltage-Controlled Oscillator (VCO). The VCO section generates a frequency-modulated signal in response to a varying input voltage, while the PLL section can lock onto an incoming FM signal and track its frequency changes. The IC also includes two different phase comparators.

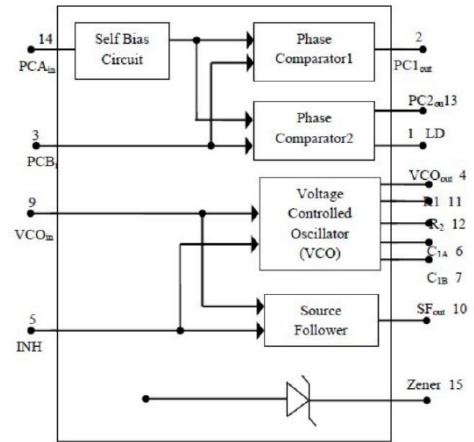


Fig. 1. Block Diagram of CD4046

II. METHODOLOGY

The experiment was carried out in two main stages: frequency modulation and demodulation using the CD4046 Phase-Locked Loop (PLL) integrated circuit. Both hardware implementation on a breadboard and simulation using PSpice software were performed to study the signal behavior and validate theoretical concepts. The following circuit diagram[2] was followed:

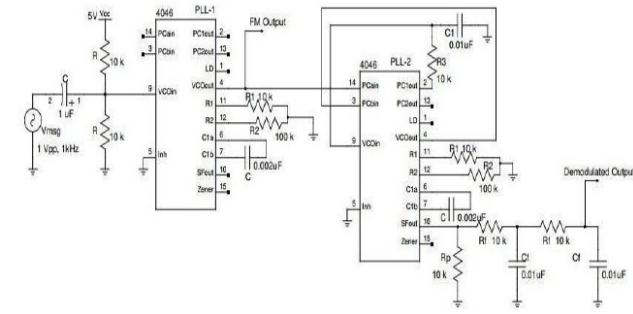


Fig. 2. Circuit Diagram

A. Frequency Modulated Signal Generation with VCO

In the modulation phase, the Voltage-Controlled Oscillator (VCO) section of the CD4046 was used to generate a frequency-modulated signal. The VCO was configured using external components—resistors R1, R2, and a capacitor C2—to set a center frequency. A 2Vp-p sinusoidal signal of 1 kHz was applied from a signal generator to the control voltage input at Pin 9 of the IC. As this input voltage varied, the VCO output at Pin 4 produced a square wave whose frequency changed in accordance with the amplitude of the input message signal. The modulated output waveform was observed and analyzed using a digital storage oscilloscope (DSO).

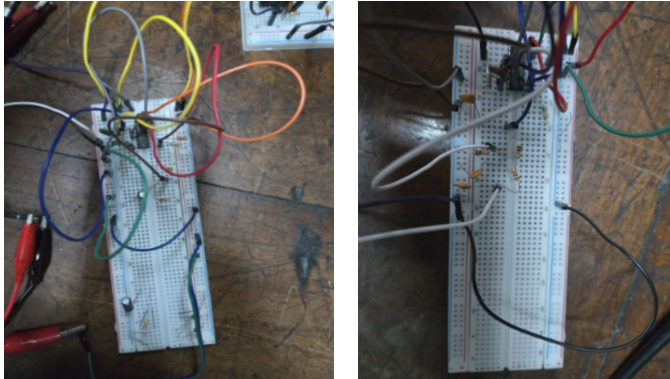


Fig. 3. Modulation and Demodulation Circuits

B. Frequency Modulated Signal Demodulation with PLL

While demodulating, the FM signal was fed into the input of a second CD4046 configured as a PLL. The phase comparator compared the incoming FM signal with the output of the internal VCO. As the PLL locked onto the frequency of the modulated signal, it produced an error voltage that tracked the frequency changes. This error voltage, which was proportional to the original message signal, was obtained at the output (Pin 10). To eliminate high-frequency noise from the comparator output, the signal was passed through a two-stage RC low-pass filter of 1.5 kHz cutoff frequency. The resulting signal was a clean reproduction of the original sinusoidal message which was observed on the oscilloscope and compared to the input. The hardware implementation is shown in figure 3.

III. RESULTS

A. Calculating Center Frequency

In a VCO, the output frequency is proportional to the control voltage input. In the absence of control voltage, the center frequency is determined by the supply voltage VCC, the externally connected resistances R1 and R2 and the capacitance C2. The values of R1, R2, C2 and VCC were as follows:

- $R_1 = 9,000 \Omega$
- $R_2 = 92,000 \Omega$
- $C = 0.002 \mu F$
- $V_{CC} = 5 V$

Then the free-running frequency/center frequency of the PLL was calculated using the equation below:

$$f_0 = \frac{0.16 \times \frac{V_{CC}}{2}}{R_1 C} + \frac{1}{R_2 C} = 27,980 \text{ Hz} \quad (5)$$

B. Modulation

In the modulation phase, when the message signal was applied to the VCO input of the CD4046, the VCO output produced a square wave whose frequency varied continuously with the amplitude of the message signal. The output waveform observed on the oscilloscope showed a square wave with varying frequency. The frequencies we observed in the modulated signal were:

- $f_1 = 27,472.53 \text{ Hz}$
- $f_2 = 28,735.63 \text{ Hz}$

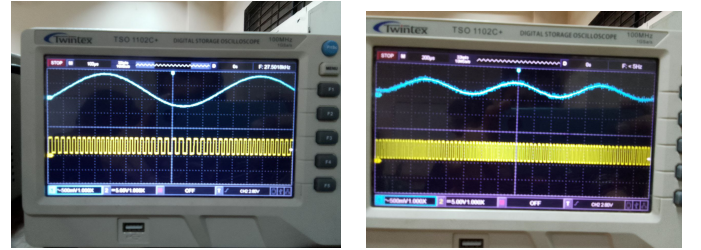


Fig. 4. Modulation and Demodulation Outputs

C. Demodulation

In the demodulation phase, the modulated FM signal was fed into the phase comparator input of a second CD4046. The internal VCO tracked the incoming signal's frequency and the resulting error voltage represented the original message signal. After passing through a two-stage RC low-pass filter to remove high-frequency ripple, the resulting waveform was observed to be a clean sinusoidal signal that closely resembled the original 1 kHz input message. The observed modulation and demodulation outputs are shown in figure 4.

The comparison between the input (modulating) signal and the recovered (demodulated) signal showed a strong match in frequency, shape, and phase (with minor delays due to filtering). This confirmed the successful recovery of the message from the frequency-modulated signal.

D. Simulation Result

Simulation results from PSpice displayed in figure 5 closely matched the hardware observations. The VCO output in simulation showed the expected frequency shifts and the demodulated output after filtering resembled the original sine wave.

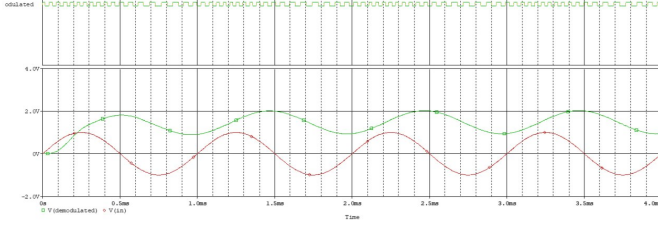


Fig. 5. Simulation Result

E. Calculating Modulation Index

$$\Delta f = \max(|f_1 - f_0|, |f_2 - f_0|)$$

$$= \max(|\{27472.53 \text{ Hz} - 27980 \text{ Hz}|,$$

$$|28735.63 \text{ Hz} - 27980 \text{ Hz}|)$$

$$\Delta f = \max(507.47 \text{ Hz}, 755.63 \text{ Hz}) = 755.63 \text{ Hz}$$

Here,

$$f_m = 1000 \text{ Hz}$$

$$\beta = \frac{\Delta f}{f_m} = \frac{755.63 \text{ Hz}}{1000 \text{ Hz}} = 0.756$$

The modulation index for this FM system is:

$$\beta = 0.756$$

IV. PROBLEMS AND PROBABLE SOLUTIONS

One of the primary limitations encountered was noise in components. Electronic devices and connecting wires inherently introduce noise into the system, which affected the clarity of the output signals. This led to minor deviations between the experimental and calculated values. However, the results were still reasonably accurate and close to theoretical expectations.

Another issue involved compatibility among various tools and hardware components. Some devices or simulation environments may not work seamlessly together, leading to integration challenges and delays in testing and troubleshooting.

Cost constraints also limited the use of higher-precision components or advanced measuring equipment.

Lastly, temperature sensitivity of electronic components was a potential concern. Components such as resistors and capacitors may behave differently with temperature variations, potentially leading to drift or efficiency loss in operation.

To reduce noise and improve stability in FM modulation/demodulation, proper grounding, decoupling capacitors and accurate component selection are essential. Using temperature-stable parts and verified simulation tools can further enhance performance and reliability.

V. COST ANALYSIS

The project was highly cost-effective, using low-cost components like the CD4046 IC, resistors, and capacitors. All equipment used was standard in educational labs. The simulation was done using freely available software. The breakdown of the costs of the components is given below:

TABLE I
COST ANALYSIS

Component	Unit	Price/Unit (Taka)	Cost (Taka)
CD 4046	2	25	50
Breadboard	2	145	290
Capacitor	10	5	50
Resistor	20	2	40
Jumper	1 set	150	150
Total			580

VI. CONCLUSION

This experiment effectively demonstrated the working principles of frequency modulation (FM) and demodulation using the CD4046 IC. Minor signal distortion and noise were observed but were successfully mitigated using low-pass filters and stable component selection.

Beyond the scope of this experiment, frequency modulation continues to play a crucial role in modern and emerging wireless communication systems. The increasing demand for higher data rates, lower latency, and better spectrum utilization is expected to drive further research in FM-based technologies. Future advancements will likely focus on enhancing spectral efficiency, developing advanced modulation schemes, and integrating FM with cognitive radio, dynamic spectrum access, and intelligent processing systems like AI, machine learning, and the Internet of Things (IoT).

As 5G and future 6G networks evolve, FM systems may be tailored to operate efficiently within specific frequency bands, ensuring robust performance even in dynamically changing environments. Additionally, power efficiency and secure communication will become key areas of focus, especially for battery-powered and mission-critical applications. Improvements in encryption, authentication, and anti-jamming techniques will further enhance the reliability of FM-based communication.

In conclusion, this experiment not only provided a strong foundation for understanding FM and PLL behavior but also highlighted the continued relevance of FM in the ever-evolving landscape of wireless communication. With ongoing innovation and integration into advanced technologies, FM remains a promising and adaptable technique for future communication systems.

REFERENCES

- [1] X. Zhihong, G. Zongqi and C. Danjiang, "The Research and Application of FM Circuit based on CD4046 in the Traffic Information Release System," 2008 International Conference on Computer Science and Information Technology, Singapore, 2008, pp. 654-657, doi: 10.1109/ICC-SIT.2008.66.
- [2] CD4046B Datasheet(<https://www.ti.com/lit/ds/symlink/cd4046b.pdf>)
- [3] B. P. Lathi, Modern Digital and Analog Communication Systems, 3rd Edition, Oxford University Press, New York, 1998.